

LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Friday January 13, 2017
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Find all integers x such that

$$\begin{cases} 2x \equiv 1 \pmod 3, \\ x \equiv 3 \pmod 4, \\ 3x \equiv 2 \pmod 5. \end{cases}$$

2. Determine the simple infinite continued fraction expansion of $\sqrt{14}$.

3. Show that for any positive integer n,

$$\sum_{d|n} \frac{n\sigma(d)}{d} = \sum_{d|n} d\tau(d).$$

(Here σ = sum of the positive divisors, τ = the number of positive divisors.)

4. Let p be an odd prime. By factorizing $x^4 + 4$, show that

$$x^4 \equiv -4 \pmod p$$

is solvable if and only if $p \equiv 1 \pmod 4$.

5. Let p be an odd prime and let r be a primitive root $\pmod p$. Show that $-r$ is also a primitive root $\pmod p$ if and only if $p \equiv 1 \pmod 4$.

6. Let $n \in \mathbb{Z}$, and let p be a prime such that $p|n^2 + n - 1$. Show that $p \equiv \pm 1 \pmod{10}$ or $p = 5$.